

## Trends in automation products

**Sensors and transmitters.** Digital technologies with smart diagnostics and wireless connectivity

**Control valves and actuators.** Higher-precision electro-pneumatics and electrohydraulics

**AC/DC drive.** Digital architecture

**Motion control systems.** Compactness, open standards, multiple-axis capability

**Recorders.** Software based with real-time analytics

**Programmable logic controllers.** Smaller form factor, open connectivity and in-built safety

**DCS/SCADA.** KPI dashboards and Web reporting features

**MMI/HMI.** Colour touch screens

— Anup Wadhwa, director, Automation Industry Association (AIA)

packaging, etc) and electrical energy management application (transmission, distribution, etc),” explain Wadhwa and Nandakumar.

There are many parts to an industrial automation and control system, which operate in multiple layers from sensors to control systems, to SCADA to ERP.

The figure below shows the various components of an industrial automation and control automation system. The field-level devices are a series of sensors and actuators controlled by intelligent logic controllers. The control network is industrial-quality Ethernet, RS-485, CAN, ModBus or other protocols. It also includes DCS/SCADA, HMI and PLC/PAC systems. The office network is dominated by the regular Ethernet network and the automation servers and PC network.

“We are seeing manufacturing equipment customers being challenged by the growing complexity of software used to control their machines,” says Tony Lennon, MathWorks industry manager for industrial automation.

“The need for better performance and efficiency and safer operation is driving this trend. Meeting these demands often means incorporating more software to control complex dynamics and provide better supervisory and fault detection logic. MathWorks helps these industrial equipment companies develop better control systems by providing modeling and simulation tools, such as MATLAB and Simulink, which reduce software development time by 50 per cent on average,” he adds.

Talking about innovations in this space, Choudhury says, “The industrial automation and control industry is characterised by smaller volumes and a huge variety of applications servicing diverse needs. Automation companies customise products for specific applications and end user requirements. Hence innovation comes more from targeted applications and less so from the application of the hottest and the newest technologies.”

“The one exception to that is on the sensor and the human-machine interface (HMI) side, where significant innovations and developments have resulted. Wide-ranging sensors for measurements as well as touch screen interfaces are bringing growth to the technology providers,” he says, adding, “Another unique characteristic of such automation is the decades of reliability and shelf life that the equipment and the resultant semiconductor parts have to endure. Automating a process or the factories is very expensive, and,

once done, it stays for several years. Hence a highly configurable and flexible system is always desired besides the reliability.”

Choudhury further states, “Another trend is energy consumption, which, in turn, reflects total operations cost. This is driving the need for more efficient motor design and deployment. For over 40 years, designers of industrial process control systems and Analog Devices Inc. have worked together to define, develop and deploy complete signal chain solutions, optimised for a wide array of applications.”

Digi International provides equipment that are useful in several levels of an industrial automation and control system.

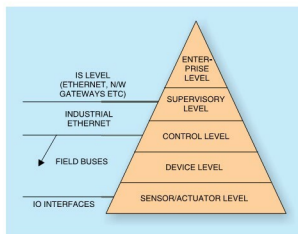
“The key market trend is to identify effective ways to increase integration across all enterprise systems, and within the industrial facility to make things more flexible so as to increase the quality of information being used to guide production,” according to Ben Tucker, sales application engineer, Digi M2M Solutions India.

## Market drivers

“Solutions capability is the key driver. Companies established in India are doing a lot of value addition in terms of application engineering and application software, including embedded software, other automation support services like consulting, project engineering, simulation, commissioning, AMC, education and training. This

has created comfort and confidence in the eco-system about sustaining the drive for modernisation and adapting to the latest technologies,” says Wadhwa.

Business in India is tightly linked to the global market—which values the ability to get information about manufacturing, delivery, forecast and quality transparently across



Network hierarchy